AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 4 · STUDENT LESSON

Python and Table Readiness

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Write the contract. Check the row. Name the denominator.

Validate structure without confusing successful execution with truth.

Keep one habit: Schema → supplied rows → missing/zero picture → complete notebook detective → independent summary.

MAIN LESSON · SECTION 1

Make every table value interpretable

> A successful Python run does not prove that the measurements are true.

Learning goals

I can specify a table **schema**.

I can check **type**s and **unit**s as well as names.

I can distinguish structural validity from truth.

Retrieve the earlier distinction

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Classification or Regression?. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from The ML Lifecycle. Give an example and a mistake that the idea helps you avoid.

Check readiness for rows, files and recovery

- What does row["used"] mean?
- Is "12" immediately usable as an integer count?
- Which error can int("twelve") raise?

My readiness explanation

My readiness explanation

Your end goal

By the end, I can explain python and table readiness using a traced example and a stated limitation.

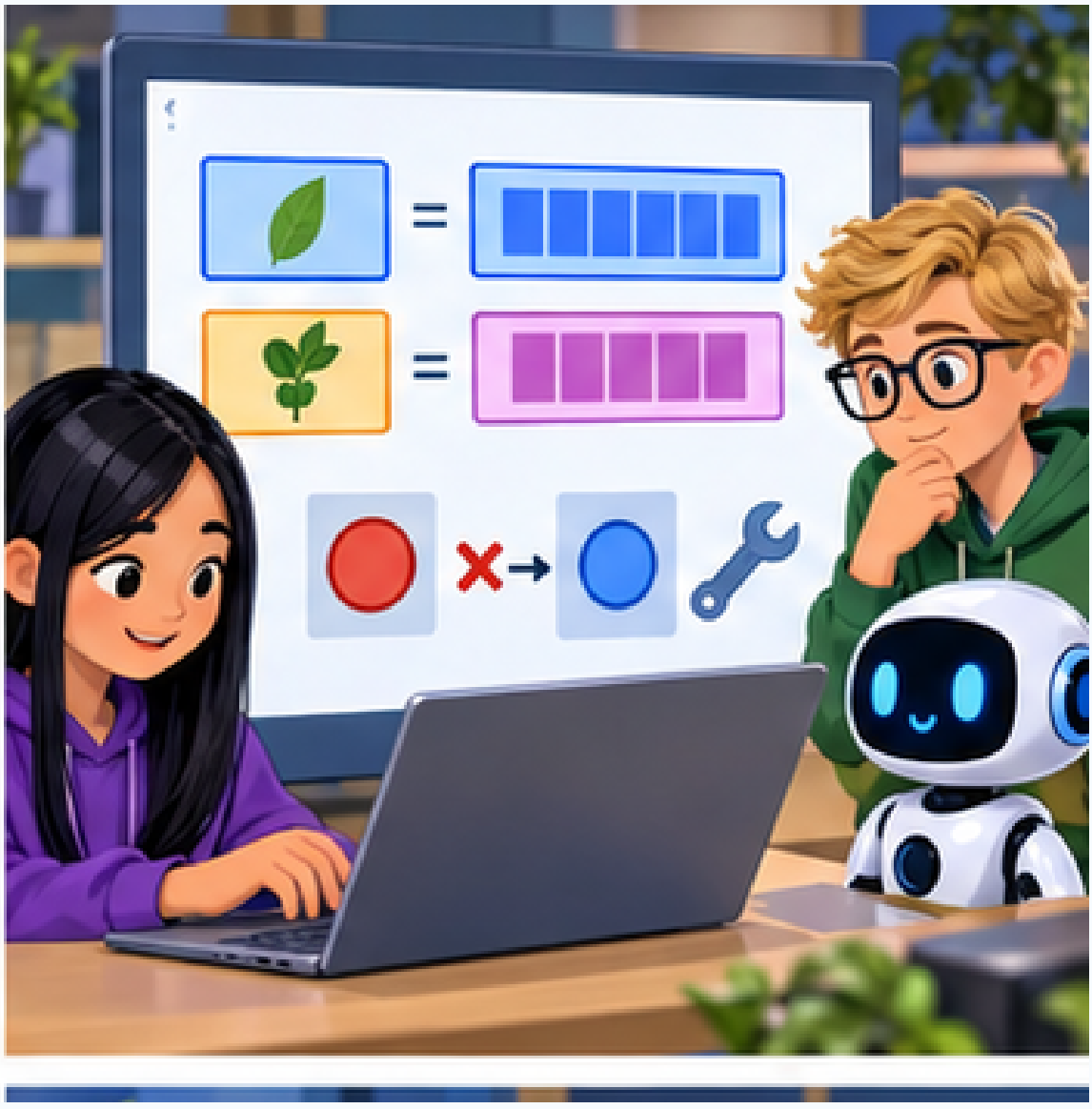
▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Python **variable**s, **type**s, **expression**s, rows, and columns give a reproducible representation of data and transformations.



LEARNING TARGET Trace mean for [4,8,12] using sum(values)/len(values). What should the program do for []? Explain why returning 0 silently is misleading.

Builder name and experiment date:

Readiness for the connected explanation

Readiness for File input and output

Readiness for Exceptions and recovery

This week's end goal

MAIN LESSON · SECTION 2

Start with one row and its written contract

> Keep the core schema check separate from optional file-reading and recovery extensions.

Main activity-session schema

A fictional classroom logs the duration of each activity. Before comparing durations, define a **schema**: activity is nonempty text, minutes is an integer from 0 through 60, and each row describes one activity session.

These decisions make data interpretable.

▼ **Optional extension · Read supplied local text**

Use a teacher-provided local sandbox folder, never a real personal **file**. Create durations.txt with two lines: 12 and 8. The goal is to read the text, convert it to integers and report the total.

Keep the original **file** unchanged for this activity.

▼ Optional extension · Parse with explicit failure status

A small parser receives duration text. Its contract permits a string representing a nonnegative integer up to 60. Invalid numeric text should return a clear **status** rather than crash or pretend that the duration was zero.

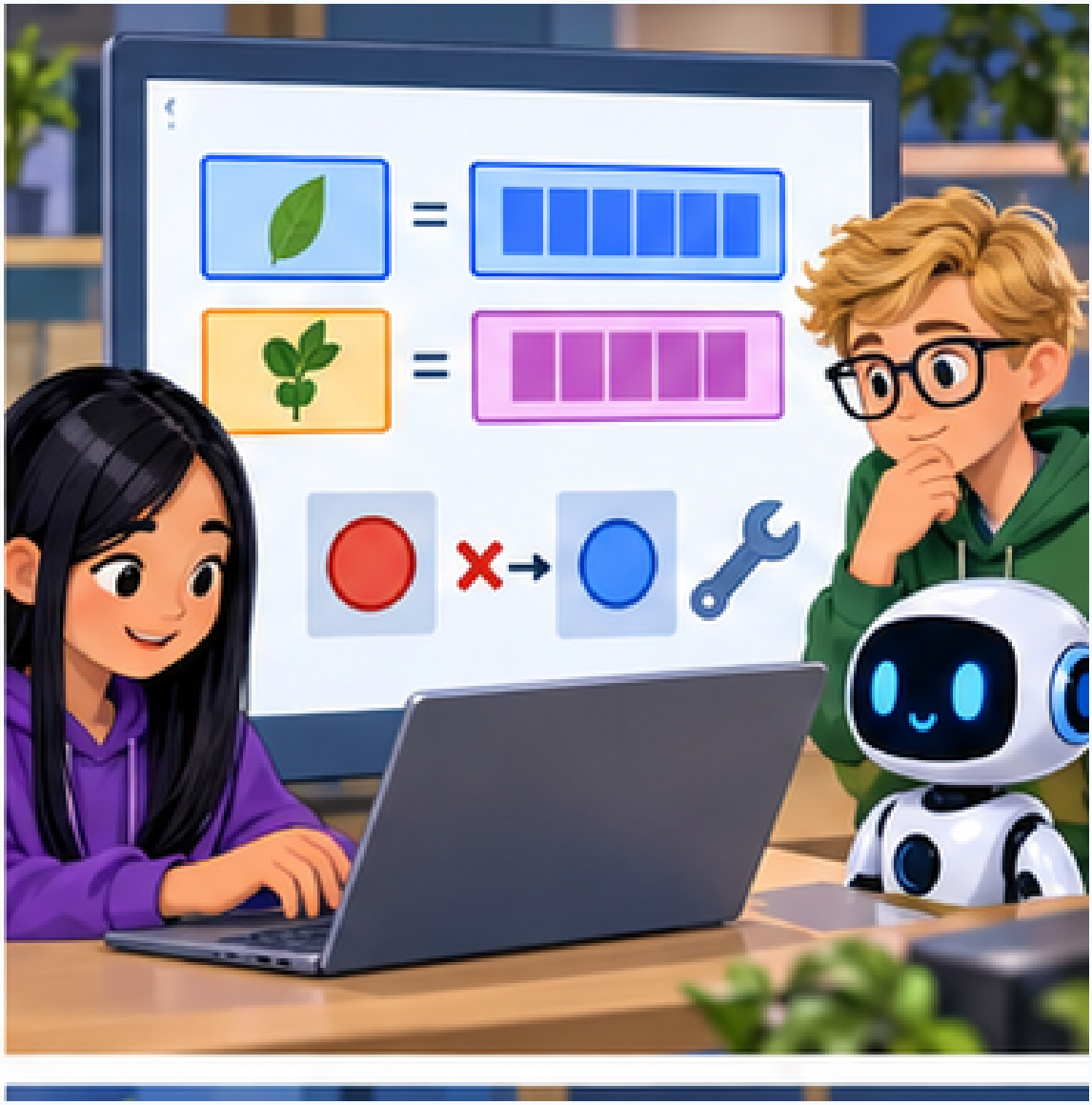
Test both the successful and failing **path**s.

▼ Earlier wording, alternatives and picture transcripts

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



MYSTERY The team must use python and table readiness to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

Apply the connected idea: File input and output

Apply the connected idea: Exceptions and recovery

MAIN LESSON · SECTION 3

Check field, type and range before using a row

> Trace the two True results and name what this small script does not check.

Read the complete schema

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
activity	Nonempty text	Names the session activity
minutes	Integer 0 through 60	Duration in minutes

Read each row from left to right and connect the evidence to the stated rule.

A **schema** is a checklist for what a row must contain. First ask whether the activity name is text and is not empty. Next ask whether minutes is an integer.

Then ask whether that integer lies between 0 and 60, including both ends. Keep these checks separate when explaining a rejection.

The number 12 and the text "12" look similar on a page, but only the number meets this lesson’s minutes rule.

The core script assumes both keys exist. A missing key raises KeyError rather than producing False. It accepts a space-only activity because its written rule checks length, not meaningful content; a stricter rule would need an explicit change. Boolean True fails the exact integer type check even though Python treats bool as an int subclass.

File reading and parsing below are extensions with different contracts, not evidence that a schema proves truth.

Run the supplied two-row validator

```
rows = [{"activity": "Reading", "minutes": 12},
        {"activity": "Sorting", "minutes": 0}]
for row in rows:
    valid = (type(row["activity"]) is str
             and len(row["activity"]) > 0
             and type(row["minutes"]) is int
             and 0 <= row["minutes"] <= 60)
    print(valid)
```

Explain exact types and inclusive limits

The worked script prints True twice. **type**(value) is int checks the exact **type** here and avoids accepting a Boolean as a count. The chained comparison checks both boundaries.

The example assumes both keys exist; next week handles missing fields. A zero duration is allowed by this written **schema**.

▼ Optional extension · Read text, convert, total

```
with open("durations.txt", "r", encoding="utf-8") as handle:
    lines = handle.read().splitlines()
values = []
for line in lines:
    values.append(int(line))
print(sum(values))
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Local file	12 then 8 on separate lines	Read as text
Conversion	"12" to 12 and "8" to 8	Integer values
Summary	12 + 8	Print 20

Read each row from left to right and connect the evidence to the stated rule.

The mode r means read. with closes the **file** when the block finishes, including when an error occurs. splitlines separates lines without retaining newline characters.

This example assumes each line is a valid integer text; an empty line or twelve raises ValueError during int conversion.

The **file** contains characters, not ready-made Python integers. Reading gives text; splitlines separates the lines; int converts each valid line into a number. The lines "12" and "8" become 12 and 8, so their total is 20.

Keep those three stages visible in your trace. A line such as "twelve" fails conversion, and a missing **file** is a different problem. Neither failure means the observed duration was zero.

▼ Optional extension · Parse and report known failures

```
def parse_minutes(text):
    try:
        value = int(text)
    except ValueError:
        return {"status": "invalid text", "value": None}
    if value < 0 or value > 60:
        return {"status": "out of range", "value": None}
    return {"status": "ok", "value": value}
```

```
for text in ["12", "zero", "0", "61"]:
    print(parse_minutes(text))
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
"12" or "0"	Conversion and range pass	ok with numeric value
"zero"	Conversion fails	invalid text with None
"61"	Integer outside range	out of range with None

Read each row from left to right and connect the evidence to the stated rule.

try covers the operation expected to fail; except ValueError handles that known conversion failure. Returning early ends the failing **path**. This contract expects a string: it does not claim to handle every possible object.

Spaces around an integer are accepted by int, and that behavior should be documented.

Trace the function as a decision **path**. First try to turn the text into an integer. If that conversion raises ValueError, return invalid text and None. If conversion succeeds, check the range.

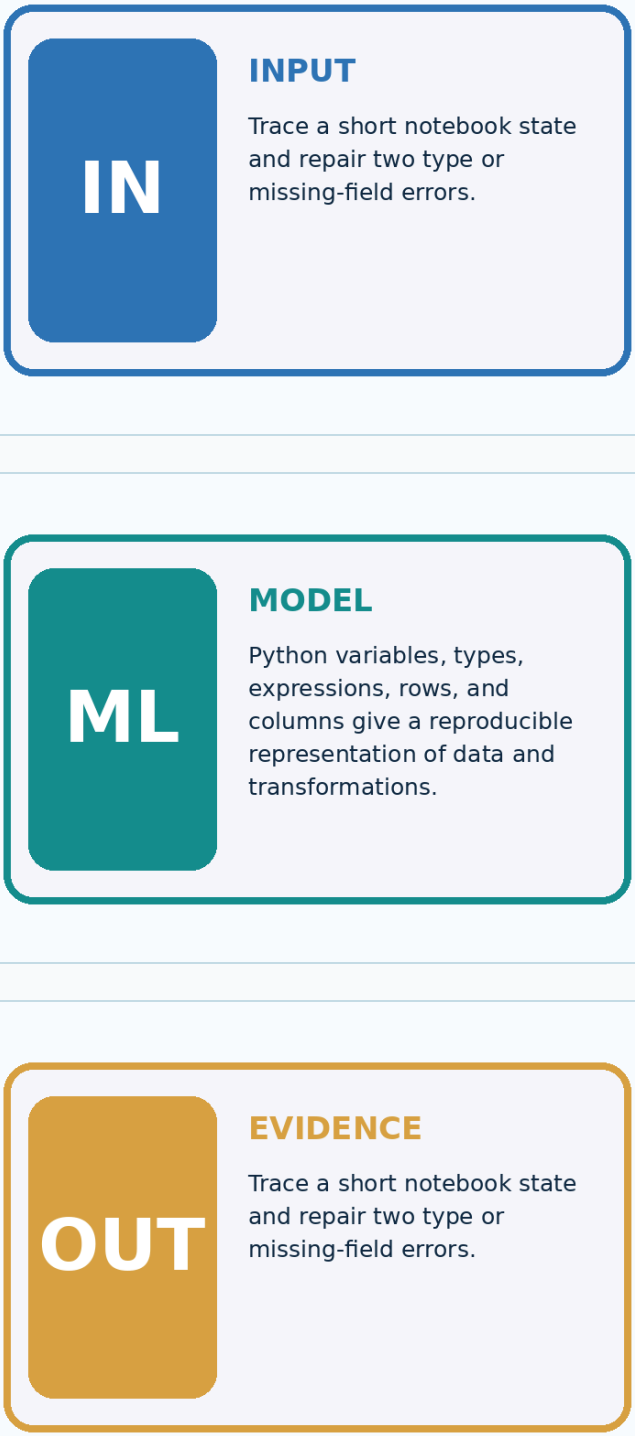
A value outside 0 through 60 returns out of range and None. Only a value passing both stages returns ok and its number. None means there is no accepted number; it must not be added to a duration total.

▼ **Earlier wording, alternatives and picture transcripts**

ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.



QUESTION Trace a short notebook state and repair two **type** or missing-field errors.

MODEL IDEA Python **variable**s, **type**s, **expression**s, rows, and columns give a reproducible representation of data and transformations.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD widths = [4.2, 5.0, 6.1] stores three decimal measurements. mean_width = sum(widths) / len(widths) evaluates to 5.1.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Check a row in separate steps

Apply the connected idea: File input and output

Separate file text from numbers

Apply the connected idea: Exceptions and recovery

Give each failure an honest result

Connect the picture and the explanation

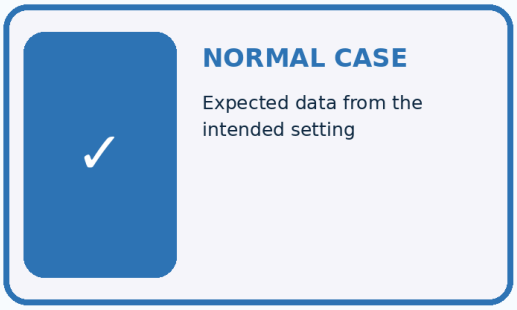
Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

MAIN LESSON · SECTION 4

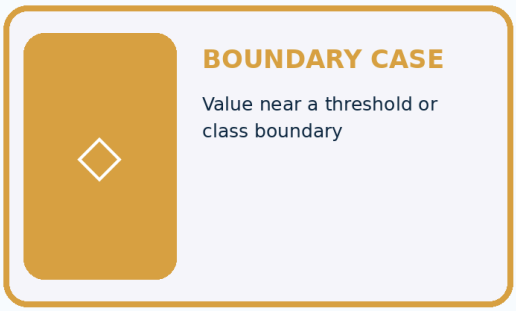
Compare normal, boundary, missing and changed-unit records

> A number can be structurally permitted and still represent the wrong quantity.

Four evaluation conditions



NORMAL Expected data and expected behavior.

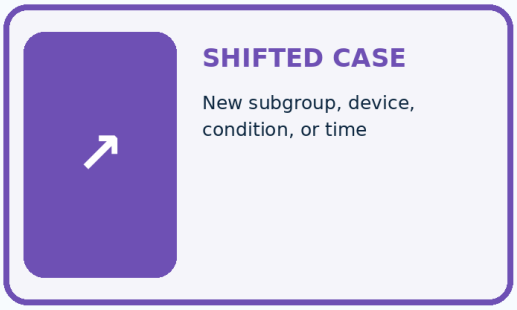


Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: S changes from 0 to None.

▼ **Check after predicting**

Prediction: Mean becomes (2+6)/2=4 cm with 2/4 measured.

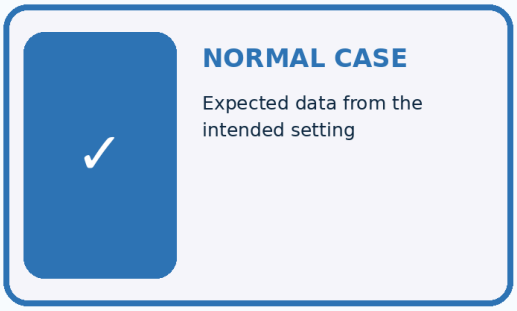
Reason: Removing a measured zero changes both the denominator and the coverage.

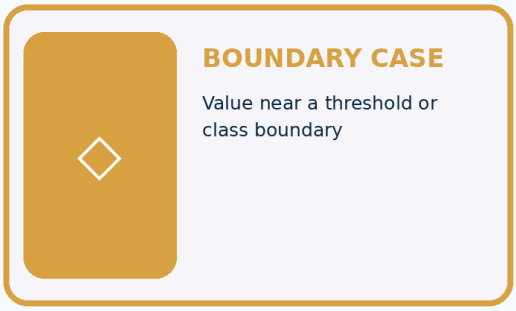
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

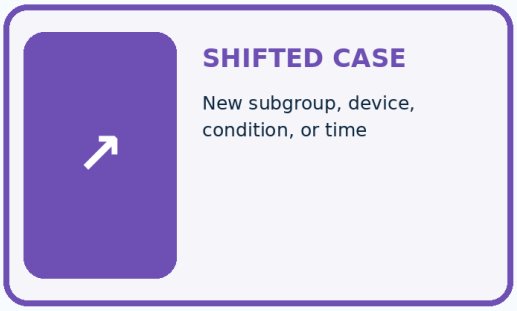
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Keep missingness separate from a measured zero

> Check the included rows before computing the denominator.

Complete supplied plant records and contract

Inspect a small table before asking Python to compute an average.

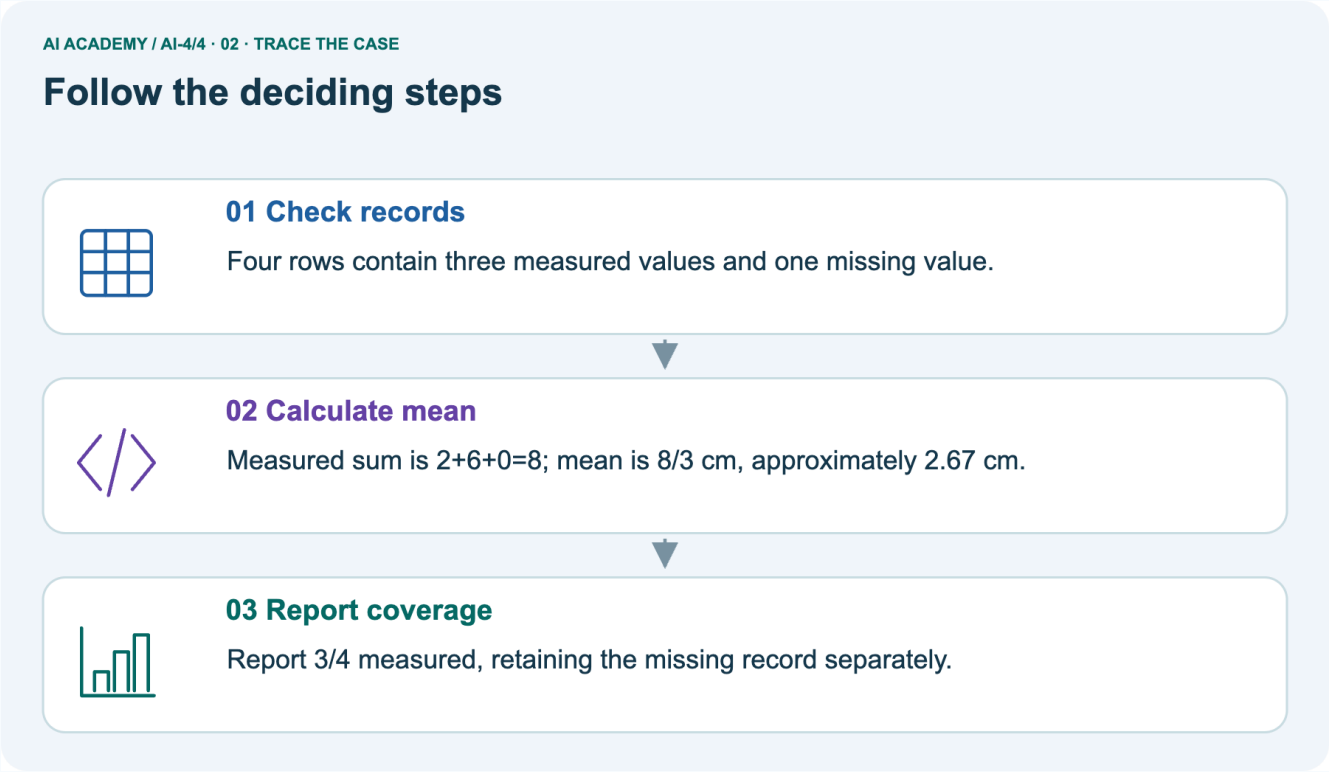
Plant heights in cm: P=2, Q=6, R=None, S=0. None means unmeasured; zero is a recorded valid height.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

This height table uses its own contract: heights are nonnegative measured numbers in cm or None for unmeasured. It is not the activity-duration schema. Exclude None explicitly, retain the observed zero and keep the original four-row denominator for coverage. The mean of available observations does not establish the mean of all four plants.

If no heights are measured, report mean unavailable and coverage 0/4; sum 0 alone does not imply measured zero height.

Trace available sum, mean and coverage



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Check records	Four rows contain three measured values and one missing value.
2. Calculate mean	Measured sum is 2+6+0=8; mean is 8/3 cm, approximately 2.67 cm.
3. Report coverage	Report 3/4 measured, retaining the missing record separately.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

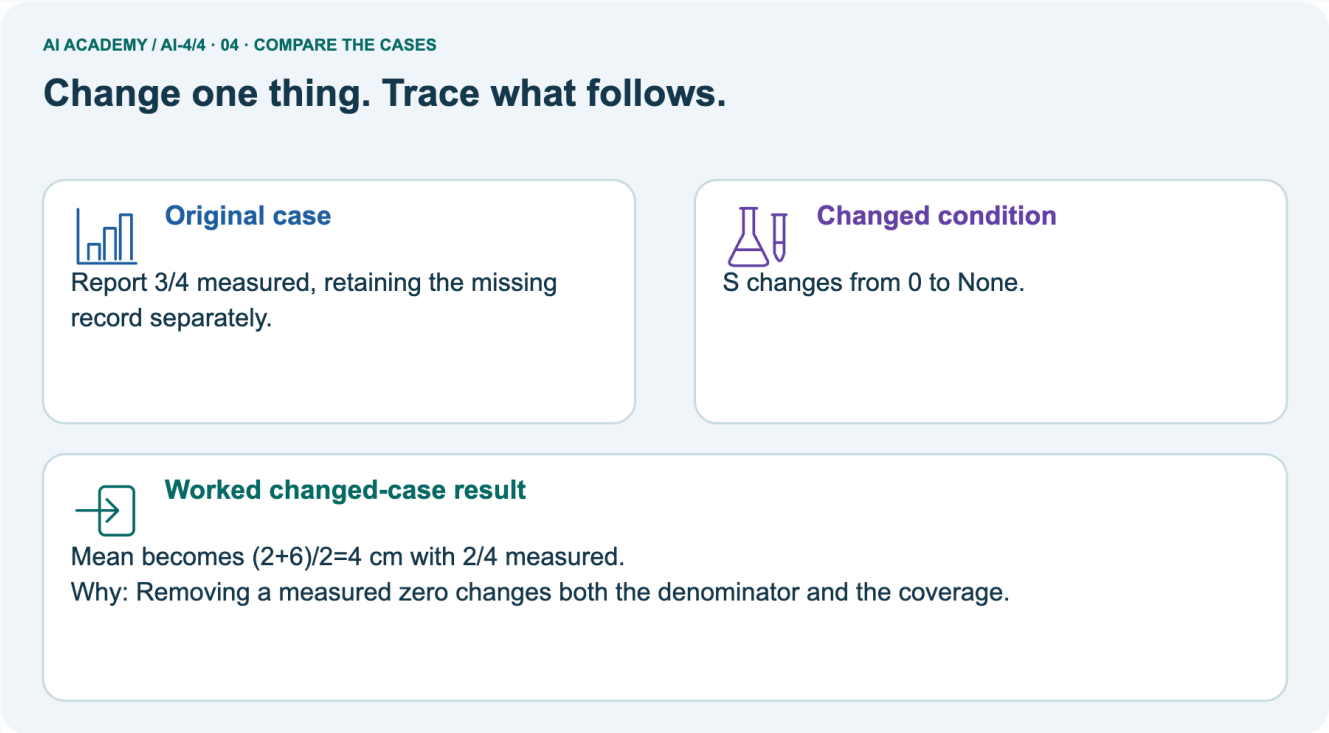
Read the picture in order. Report 3/4 measured, retaining the missing record separately. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict zero becoming missing

- Change one thing: S changes from 0 to None. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the limit and pictured decision

- The measured-row mean may not represent the missing plants; investigate why those readings are absent.

- Read the labelled picture: Why should the missing value not silently become zero? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Additional schema and extension predictions

Would minutes of 61, "12" and 12 each pass? Explain the type distinction.

If the file contains 0, 3 and 7 on separate lines, predict the values and total.

Predict statuses and values for "-1", " 5 " and "3.5".

Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 4 • WORKED EVIDENCE

1. widths = [4.2, 5.0, 6.1] stores three decimal measurements.
2. mean_width = sum(widths) / len(widths) evaluates to 5.1.
3. Adding 'unknown' to the numeric list causes a type problem.
4. Clean or explicitly encode values, rerun top to bottom, and record the result.

STEP 1 widths = [4.2, 5.0, 6.1] stores three decimal measurements.

STEP 2 mean_width = sum(widths) / len(widths) evaluates to 5.1.

STEP 3 Adding 'unknown' to the numeric list causes a **type** problem.

STEP 4 Clean or explicitly encode values, rerun top to bottom, and record the result.

CHECK Verify the denominator, **unit**, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Apply the connected idea: File input and output

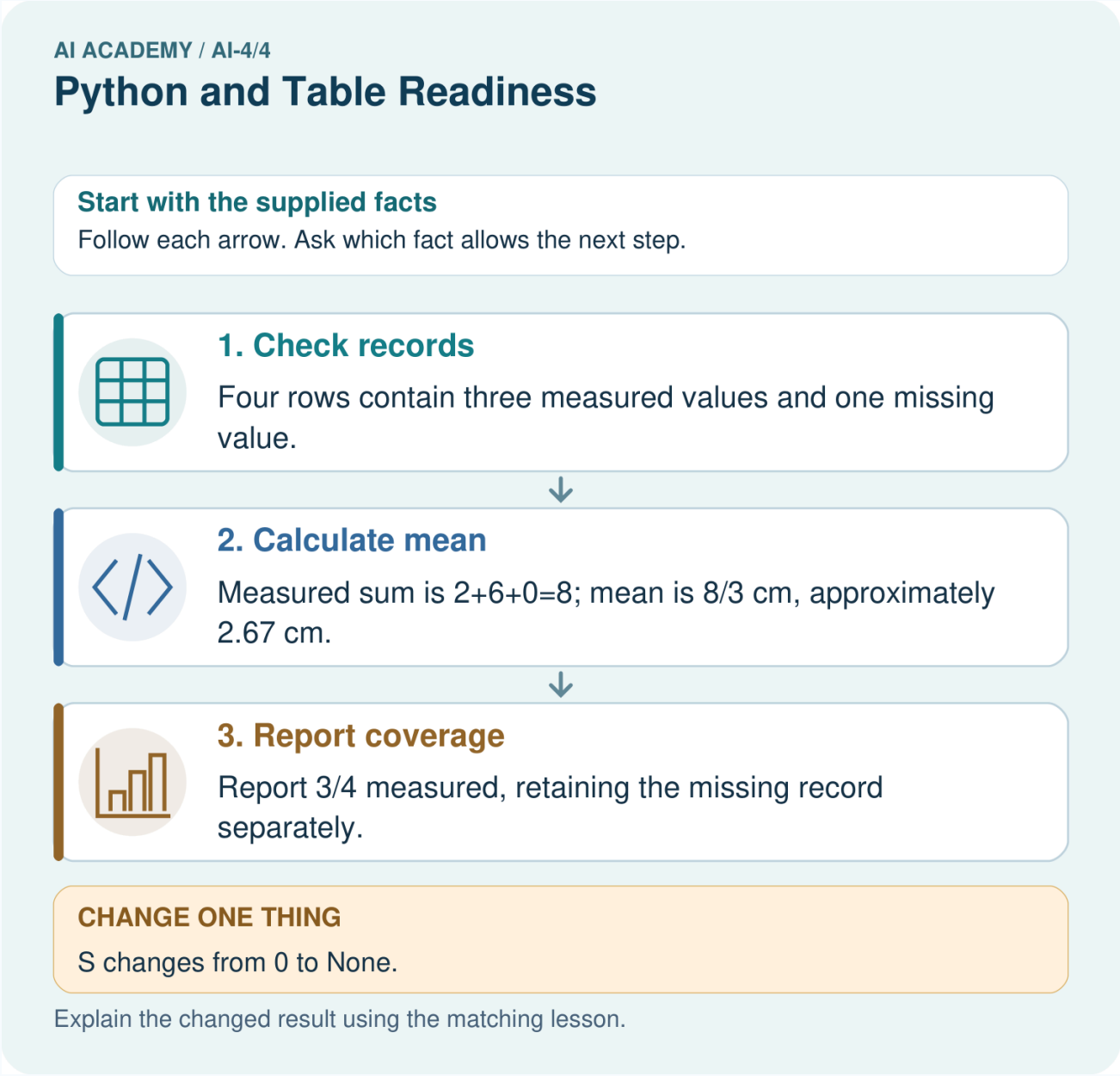
Apply the connected idea: Exceptions and recovery

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Python and Table Readiness

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find the first unsupported assumption

> Passing a structural check is different from verifying an observation.

Core validity-versus-truth question

- A row can satisfy every **schema** rule and still be a wrong **observation**. **Schema** validation checks our stated structure and **constraint**s, not whether someone measured honestly or selected a representative sample.

> **A row says Reading, 12 and passes. Does that establish the session really lasted 12 minutes?**

Challenge notebook order assumptions

- Diagnose one cause and name the cleanest next experiment:

CLAIM TO CHALLENGE If notebook cells eventually run, execution order and **data type**s do not matter.

▼ Optional file and recovery diagnosis

File reading does not establish that the values are valid durations. After conversion, apply the **schema** and record any rejected lines. Do not upload personal records to an external tool for this exercise.

The file contains twelve on one line. Why should the program not silently turn it into zero?

Recovery must fit the task. A caller may ask for corrected **input** or exclude the row with a recorded reason. It must not use None as a number or claim that rejected records were successfully **process**ed.

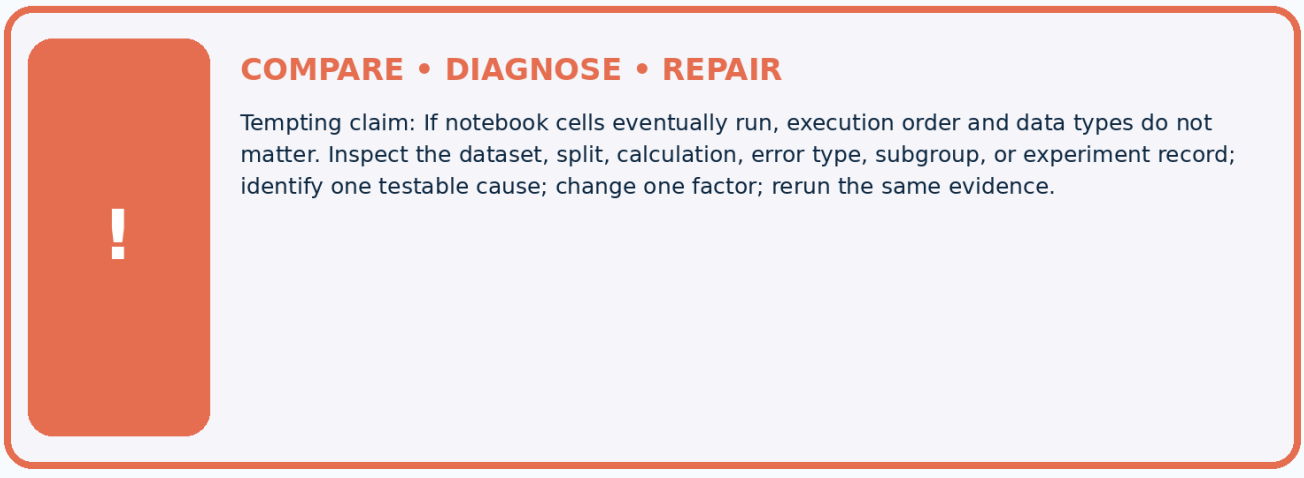
A batch has 8 valid and 2 invalid lines. What should its report include?

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.



Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

A closer explanation

Apply the connected idea: File input and output

Apply the connected idea: Exceptions and recovery

MAIN LESSON · SECTION 8

Practise inclusive limits and explicit unit conversion

> Use the written contract before accepting any numeric result.

Attempt W2 with the schema beside you

Use workbook W2 for the connected example “Rows columns and **schema**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check whether a successful run proves truth

- Use the worked case for Rows columns and **schema**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Repair a seconds/minutes mismatch

If one row records 120 seconds in the minutes field, a simple numeric total becomes misleading. Correct the source or explicitly convert and document it; renaming a column alone does not convert values.

Explain the missing-value picture

- Why should the missing value not silently become zero? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Reserve the individual task

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Optional file and parser practice**

If Python cannot find durations.txt, it raises **File**NotFoundError. Check the current working folder and exact **file**name.

Do not switch to write mode to solve a missing **input**: opening an existing **file** with w would replace its contents.

A bare except that returns 0 hides unrelated programming errors and corrupts missing values. Catch the expected **exception**, preserve an error **status** and keep the unknown value as None.

If **file** reading is added, handle **File**NotFoundError separately with an actionable message.

Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A successful run proves the conclusion

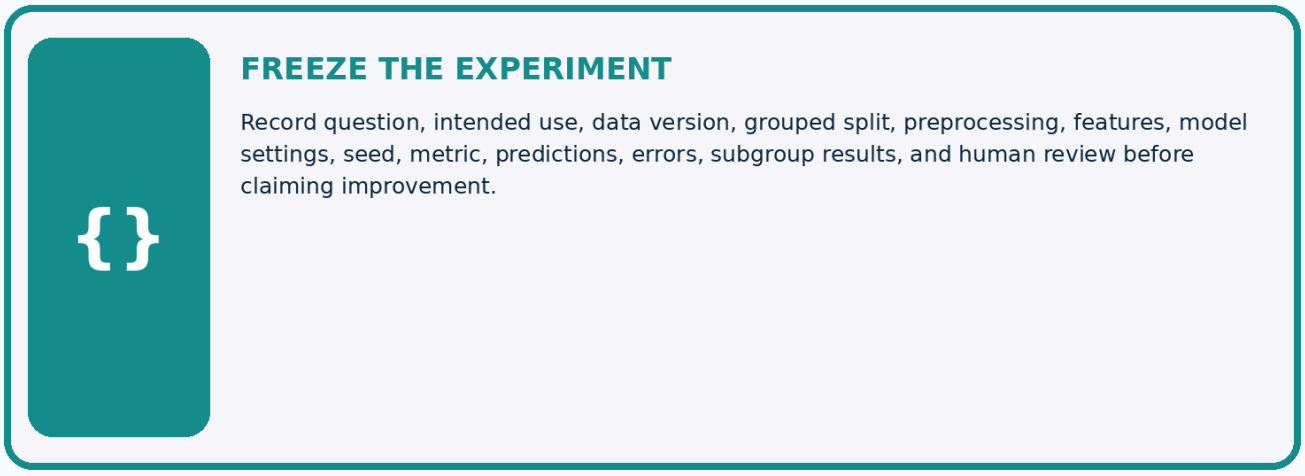
Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A successful run proves the conclusion

▼ **Earlier wording, alternatives and picture transcripts**

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + **unit**s + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Apply the connected idea: File input and output

Apply the connected idea: Exceptions and recovery

Practise with less help: Classify records

Check your reasoning

Practise the connected topic: File input and output

Practise the connected topic: Exceptions and recovery

Explain the picture

MAIN LESSON · SECTION 9

Investigate a complete notebook state and two filters

> All inputs needed for the promised detective activity are supplied here.

Complete five-variable, filter and error fixtures

Notebook Detective fixture	Predict and explain
Run in order: counts=[2,0,6]; total=sum(counts); n=len(counts); mean=total/n; observed=n	Predict all five variables; then repeat from a clean state.

Notebook Detective fixture	Predict and explain
Rows: A={activity:Reading,minutes:12}; B={activity:Sorting,minutes:0}; C={activity:Quiz,minutes:61}; D={activity:Chat,minutes:text "12"}	Filter 1: keep rows whose minutes is an exact int in 0..60. Filter 2: among those, keep minutes>0. State retained IDs.
Broken expression 1: 12 + "8"; supplied source says both are minute counts	Explain TypeError, convert the known text count intentionally, then check duration range.
Broken expression 2: {"activity":"Reading"}["minutes"]	Explain KeyError; request the missing observation rather than invent zero.
Normal/boundary/missing/shifted probes	A valid 12; B valid zero; missing minutes requires correction; recorded 120 seconds must be documented as 2 minutes before applying the minute contract.

Predict, run cleanly and explain repairs

1. Predict before running. Keep each broken expression in a separate attempt so the first error does not hide the second. Save raw rows and rejection reasons. Filter rules here are supplied explicitly: they check the minutes field on these complete rows, not every possible activity constraint. Changing notebook cells requires rerunning dependent cells from the start. Record traced rather than executed outcomes honestly.

   Keep the independent three-value mean task unreviewed until the individual attempt.

Keep evidence and actual execution status

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Notebook Detective: predict five variable values, trace two table filters, repair two errors, and rerun from a clean state.

1. Write the hypothesis and expected **evidence** before running.

2. Notebook Detective: predict five **variable** values, trace two table filters, repair two errors, and rerun from a clean state.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: specify a summary with missing information

> Keep calculation, coverage and unsupported claims separate.

Fresh supplied task and scope

Supplied values are3,7,14. Compute their mean. What should a function report for an empty list? How should it handle a missing fourth record?

Use the three supplied values as the observed records. Treat the missing fourth record as unknown, not zero. State the policy and keep observed-count and total-record-count separate. For an empty collection, explain the status before attempting division; no new measurement is supplied.

First attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after attempting

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Changed independent case

Week 4 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Trace Python: lengths = [4, 7, 9]; selected = lengths[1]; total = selected + 3. Give selected and total.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Create a tiny five-row plant table with explicit column names, **type**s, **unit**s, and a reproducible summary calculation.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

Trace mean for [4,8,12] using sum(values)/len(values). What should the program do for []? Explain why returning 0 silently is misleading.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Keep a reproducible table contract with the project

> The next lesson can inspect this table without guessing its units.

Create the five-row artifact

THIS WEEK'S CHECKPOINT Create a tiny five-row plant table with explicit column names, **type**s, **unit**s, and a reproducible summary calculation.

For the five-row project table, state one row's meaning, required names, exact types, units, allowed ranges and missing-value policy. Keep IDs and raw observations. Report included count, total count and the denominator used by each summary; keep rejected rows and reasons. A validator checks the stated contract, not measurement truth, representativeness or model quality.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Record evidence and human limits

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom proto**type** for medical, pesticide, agricultural, or other consequential decisions.

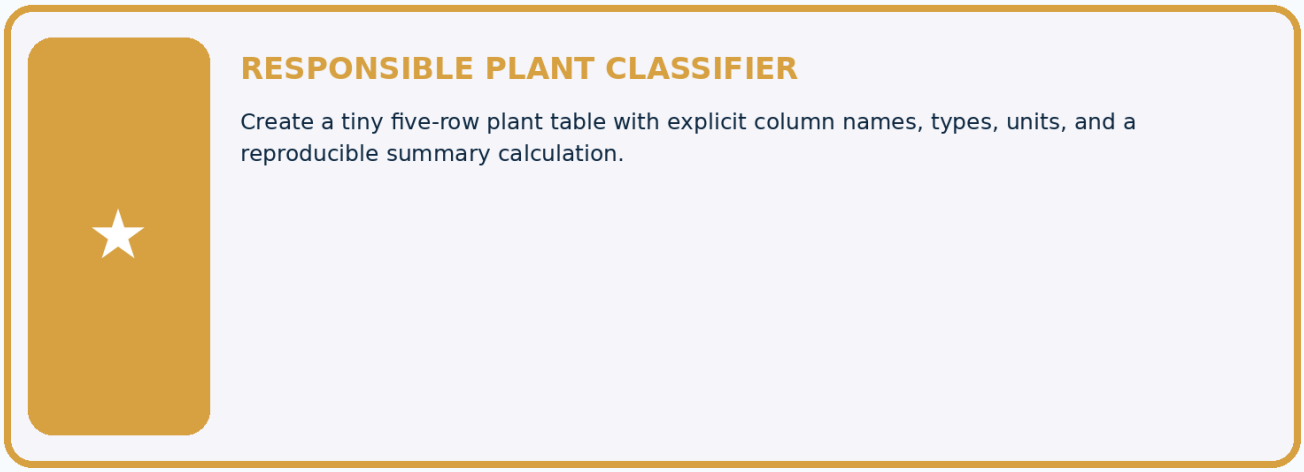
My **evidence**-linked project decision:

▼ **Earlier wording, alternatives and picture transcripts**

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

MAIN LESSON · SECTION 16

Finish with a contract and an honest denominator

> Explain why valid structure does not prove a true observation.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Specify a table **schema**

> **My deciding step, evidence and one remaining question**
>
> ---
>
> ---
>
> ---
>
> ---

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
variable	name bound to a value	
data type	kind of value and valid operations	
dataframe	table represented in code	
expression	code that computes a value	

Closer-explanation terms

Word	Meaning
Schema	Rules for the fields and values in records
Type	A category such as integer or text
Unit	The quantity scale attached to a number

Word	Meaning
Constraint	A rule a valid record must satisfy

Extension vocabulary

Term	Meaning
File	Stored data with a name and location
Path	A location used to find a file
Encoding	A rule for representing text as bytes
Context manager	A with block that manages a resource such as closing a file

Extension vocabulary

Term	Meaning
Exception	A signal that an operation could not complete normally
Handler	Code that responds to a specified exception
Recovery	A defined next step after a failure
Status	A label stating the outcome of a processing step

Earlier wording, alternatives and picture transcripts

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.


1
VARIABLE
name bound to a value
2
DATA TYPE
kind of value and valid operations
3
DATAFRAME
table represented in code
4
EXPRESSION
code that computes a value

MISCONCEPTION TO REPAIR If notebook cells eventually run, execution order and **data type**s do not matter.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Terms for File input and output

Terms for Exceptions and recovery

▼ **Extra challenge · Build a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **variable**, **data type**, **dataframe**, **expression**.
- CALCULATE Reproduce the worked **evidence** with **unit**s and denominator.
- TRACE Explain the data, code or decision **path**.
- DIAGNOSE If notebook cells eventually run, execution order and **data type**s do not matter.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, limitation, and human responsibility.

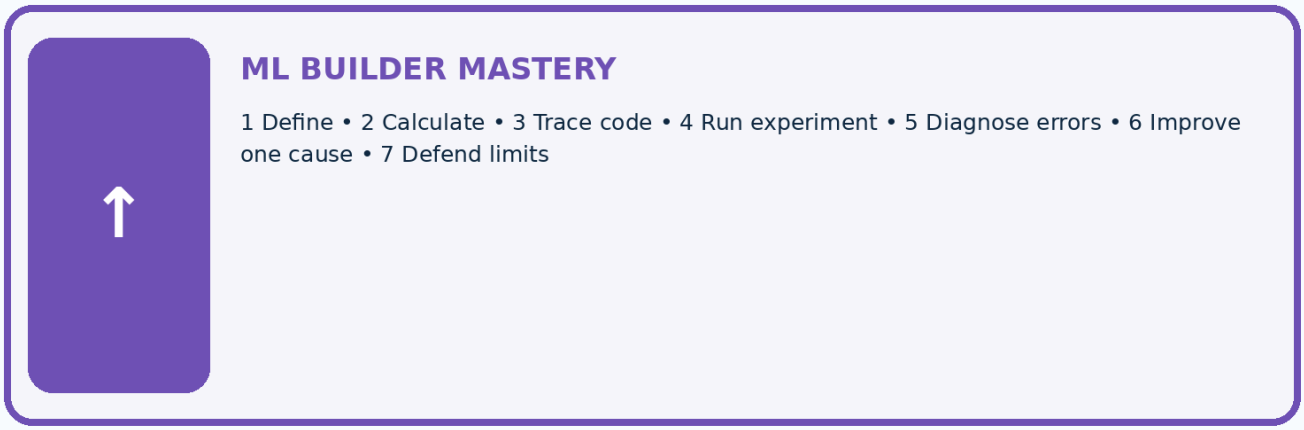
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ **Earlier wording, alternatives and picture transcripts**

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



▼ **Reference · Inspect your evidence**

REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the supplied independent observed-values and missing-record task for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.
- Independent **evidence** and defense:
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Trace a short notebook state and repair two **type** or missing-field errors.

NEXT BRIDGE Carry the experiment **evidence** into: Inspect Before Modeling.

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 13

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

Python lengths=[4,6,8]; total=sum(lengths); mean=total/len(lengths).

Trace and handle an empty list.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ **Check after trying**

Total=18, count=3, mean=6.0. Guard the empty case against division by zero; no mean exists without **observation**s.

Evidence or stage	Value or result
lengths	[4,6,8]
sum	18
count	3
mean	6.0
empty list	No mean, guard division by zero

▼ Earlier wording, alternatives and picture transcripts

Week 4 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Trace the calculation and its empty case

Python lengths=[4,6,8]; total=sum(lengths); mean=total/len(lengths).

Trace and handle an empty list.

Total=18, count=3, mean=6.0. Guard the empty case against division by zero; no mean exists without **observation**s.

▼ Reference · Earlier route checklist

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Specify a table schema	Explain the deciding step, show a trace or test, and state one limit.
Check types and units as well as names	Explain the deciding step, show a trace or test, and state one limit.
Distinguish structural validity from truth	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Inspect a small table before asking Python to compute an average.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Inspect a small table before asking Python to compute an average.
- Which labelled step matches this detail? Report 3/4 measured, retaining the missing record separately.
- 1. Check records
- 2. Report coverage
- 3. Calculate mean
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Report coverage Report 3/4 measured, retaining the missing record separately. Read the picture in order. Report 3/4 measured, retaining the missing record separately.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: S changes from 0 to None. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. Mean becomes (2+6)/2=4 cm with 2/4 measured.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Mean becomes (2+6)/2=4 cm with 2/4 measured. Removing a measured zero changes both the denominator and the coverage.

Repair and keep the record

Review Classification or Regression? (ai-4/3). Goal: Identify a task from the meaning of its target. Short practice: Did regression MAE improve when the boundary changed?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review The ML Lifecycle (ai-4/1). Goal: Distinguish **fitting** a rule from applying a fixed rule. Short practice: Should the team silently retune on the changed test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

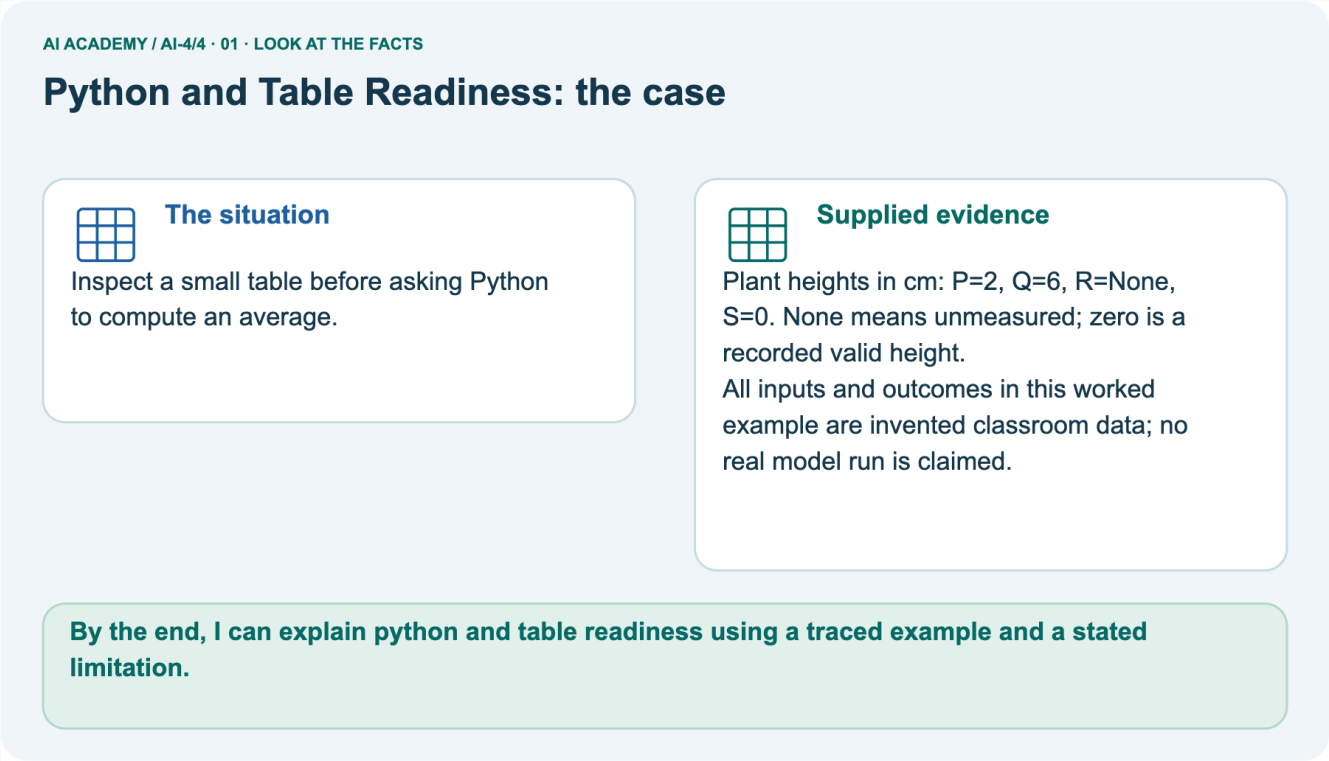
▼ **Reference · The six-picture journey**

REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Python and Table Readiness: the case



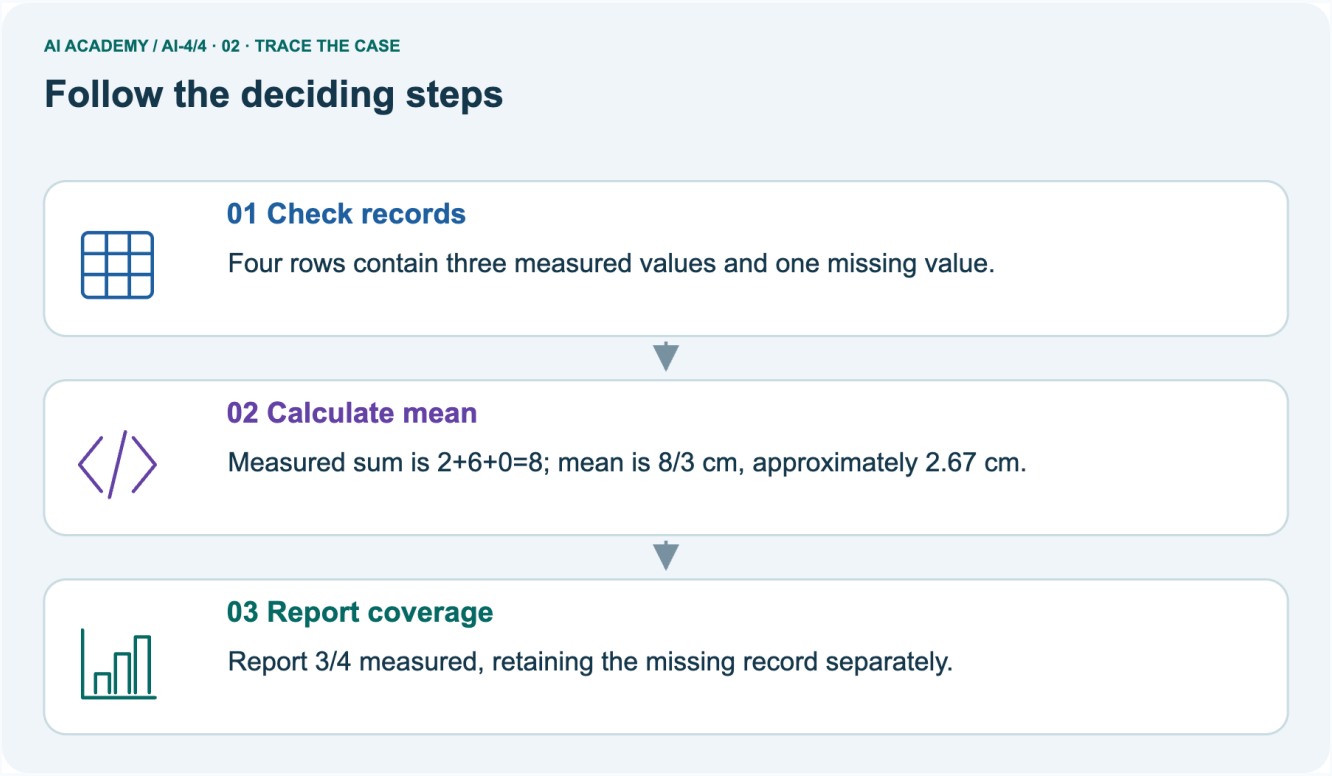
Follow the labelled picture with the matching explanation. Open the vector diagram

Inspect a small table before asking Python to compute an average. Supplied facts: Plant heights in cm: P=2, Q=6, R=None, S=0. None means unmeasured; zero is a recorded valid height.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

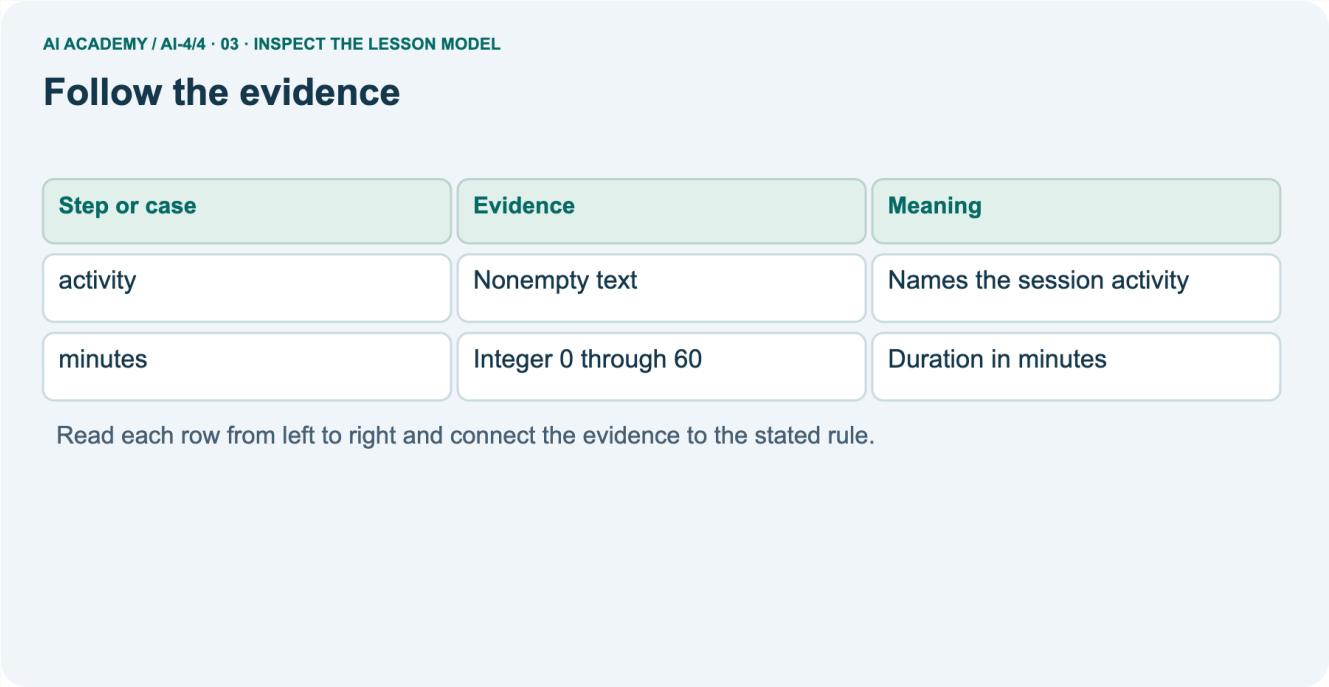


Follow the labelled picture with the matching explanation. Open the vector diagram

Check records: Four rows contain three measured values and one missing value. → Calculate mean: Measured sum is 2+6+0=8; mean is 8/3 cm, approximately 2.67 cm. → Report coverage: Report 3/4 measured, retaining the missing record separately.

Point to each numbered stage and explain which recorded **input** it uses.

03. Follow the evidence

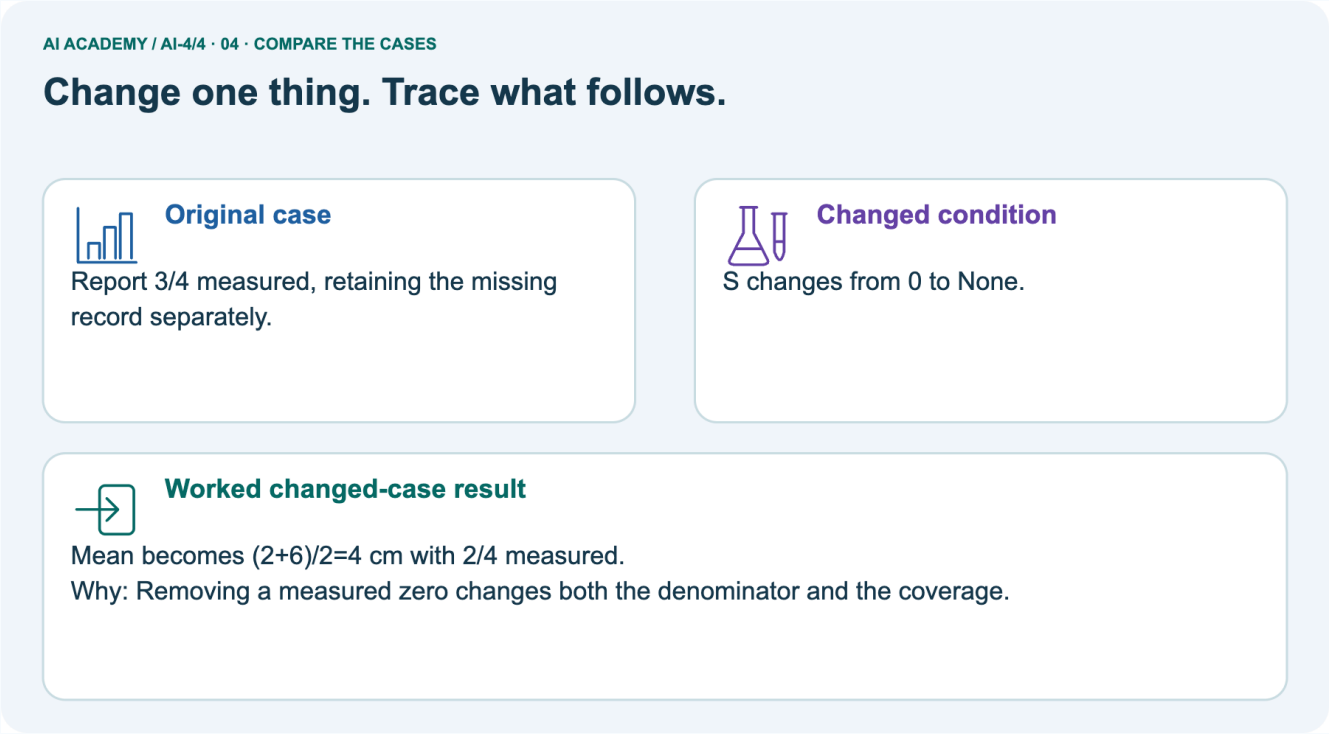


Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. activity | Nonempty text | Names the session activity. minutes | Integer 0 through 60 | Duration in minutes

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



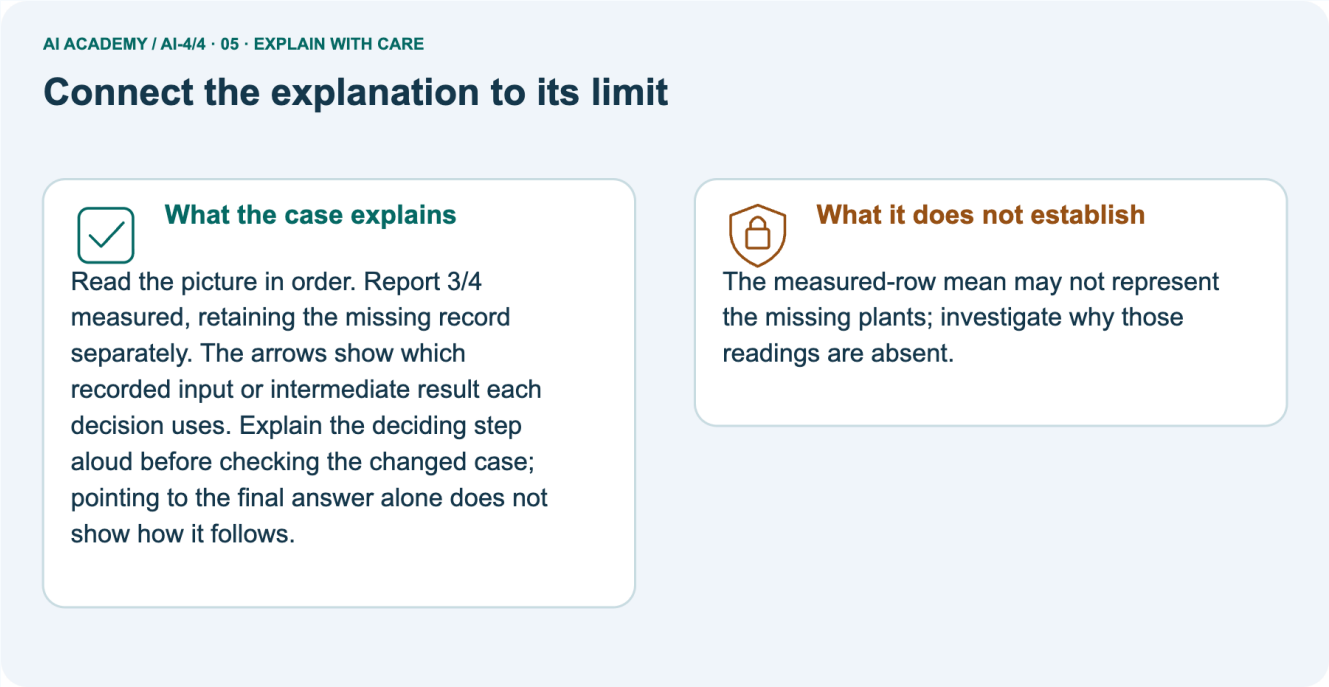
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Report 3/4 measured, retaining the missing record separately. Changed condition: S changes from 0 to None. Expected result: Mean becomes (2+6)/2=4 cm with 2/4 measured.

Reason: Removing a measured zero changes both the denominator and the coverage.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

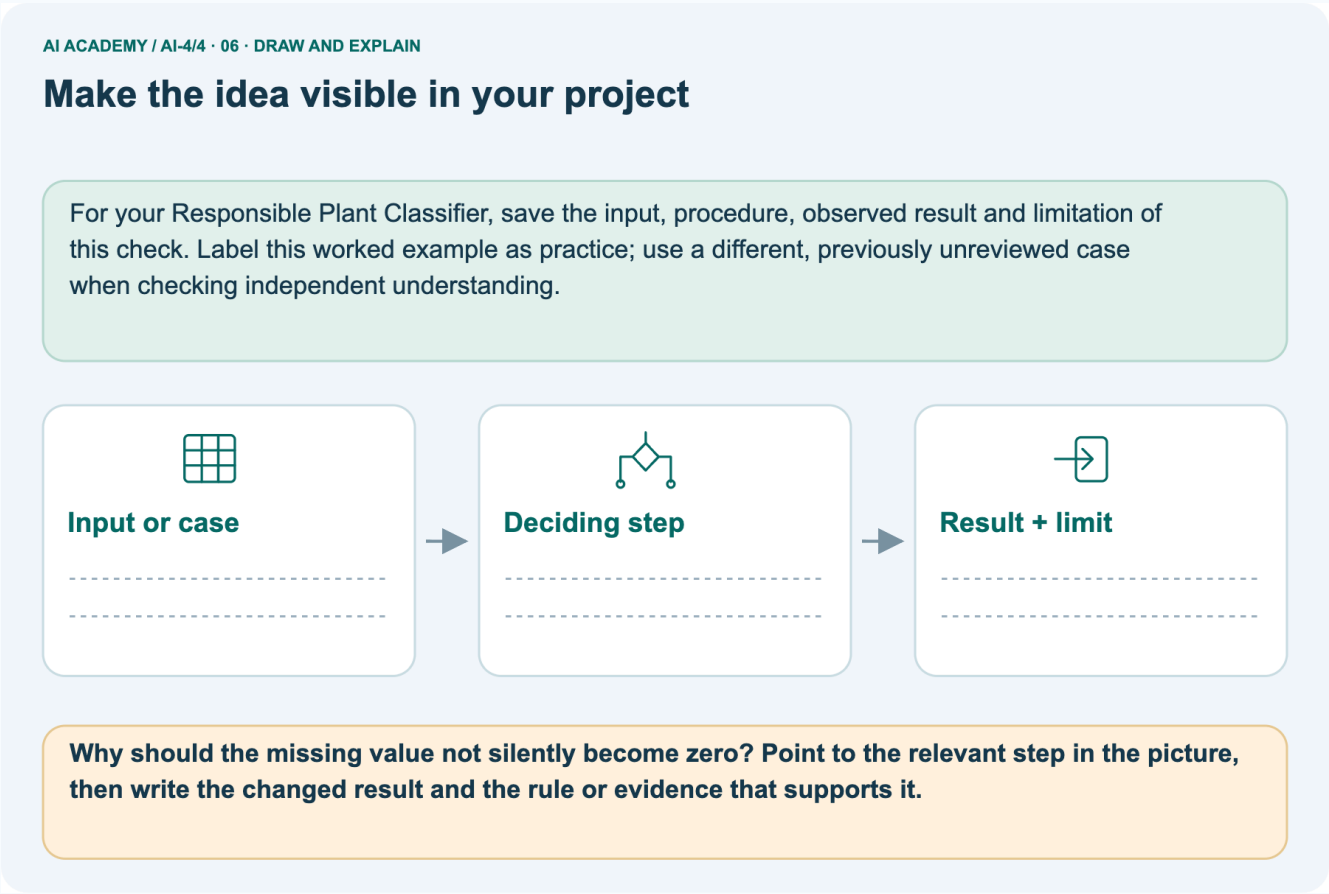
Read the picture in order. Report 3/4 measured, retaining the missing record separately. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: The measured-row mean may not represent the missing plants; investigate why those readings are absent.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why should the missing value not silently become zero? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Python and Table Readiness: the case

02. Follow the deciding steps

03. Follow the evidence

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.